dictionary. If the contact is not in the list, a popup is shown to the user, who can then decide to save it. In that scenario, to save the new contact another HTTP call is made to the web server, and at the same time the dictionary is updated with the new value (although without going through the server).

So far for the application design: figure 4.2 shows a possible architecture for our minimum viable product. Now you just need an efficient way to browse a contacts list and check if an email belongs to it.

Looking through a list for a certain entry is a common problem in computer science, known as the *dictionary problem*.

4.2 Alternatives to Implement a Dictionary

The name shouldn't be surprising, it's exactly like when you need to look up a word in a dictionary (and here by dictionary I mean one of those old gigantic paper books that have been almost entirely replaced by online dictionaries and search engines) or even in a phone book (which neither had better luck after the 3rd computer revolution).

To recap, our contacts web app needs to:

- Download the list of contacts from a server;
- Create a local copy for fast lookup/storage;
- Allow looking up for a contact;